# C# 1D Array Operations

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**Note:** Each example is a separate complete C# program. For insertion examples, the array has one extra slot. Deletion examples use `size` to track active elements.

## 1. Array Traversal

```
using System;

class Program {     static void Main()     {
int[] numbers = { 10, 20, 30, 40, 50 };

        for (int i = 0; i < numbers.Length; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 2. Array Insertion at the End

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;

        int newValue = 60;

        numbers[4] = newValue;

        for (int i = 0; i < 5; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 3. Array Insertion at the Beginning

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;

        int newValue = 5;

        for (int i = 3; i >= 0; i--)
{             numbers[i + 1] =
numbers[i];         }

        numbers[0] = newValue;

        for (int i = 0; i < 5; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 4. Array Insertion at a Specific Position

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;

        int position = 2;
int newValue = 35;

        for (int i = 3; i >= position - 1; i--)
{             numbers[i + 1] = numbers[i];
}

        numbers[position - 1] = newValue;

        for (int i = 0; i < 5; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 5. Array Deletion at the End

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;

        int size = 4;

        // Delete at the end
size--;

        for (int i = 0; i < size; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 6. Array Deletion at the Beginning

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;

        int size = 4;

        // Delete at the beginning
for (int i = 0; i < size - 1; i++)
{             numbers[i] = numbers[i + 1];
}

        size--;

       for (int i = 0; i < size; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 7. Array Deletion at a Specific Position

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;
numbers[3] = 40;

        int size = 4;
int position = 3;

        // Delete at a specific position         for
(int i = position - 1; i < size - 1; i++)         {
numbers[i] = numbers[i + 1];         }

        size--;

        // Display the array
for (int i = 0; i < size; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 8. Array Updation

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[4];

        numbers[0] = 10;
numbers[1] = 20;
numbers[2] = 30;         numbers
[3] = 40;

        int position = 3;
int newValue = 35;

        // Update the array
numbers[position - 1] = newValue;

        for (int i = 0; i < numbers.Length; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

## 9. Array Searching (Linear Search)

```
using System;

class Program {     static void Main()     {
int [] numbers = { 10, 20, 30, 40, 50 };

        int searchValue = 20;
bool found = false;

        for (int i = 0; i < numbers.Length; i++)         {
if (numbers[i] == searchValue)             {
Console.WriteLine("Value found at index " + i);
found = true;                 break;             }         }

        if (found == false)
{
            Console.WriteLine("Value not found");
        }
    }
}
```

## 10. Array Reset

```
using System;

class Program {     static void Main()
{          int[] numbers = { 10, 20, 30, 40
};

        for (int i = 0; i < numbers.Length; i++)
{             numbers[i] = 0;         }

        for (int i = 0; i < numbers.Length; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```